USER MANUAL

Mini OS NeOS v2.0

Project	Python-based command-line operating system simulator
Document	User guide for setup, login, commands, and shutdown
Version	2.0

Prepared by

Neo Sebastiene C. Halaba
Kody Benz O. Reyes
Paul Nazarene S. Jordan
Daphnie Claire A. Montalban
Kym Niecoole R. Jabican
Aaron John Demakiling

MiniOS NeOS is an educational simulator. It is not a bootable operating system and does not control actual operating-system

1. **Introduction**

Mini OS NeOS is a lightweight command-line program written in Python for learning basic operating-system concepts. After signing in, a user can manage files and directories in a dedicated workspace, inspect simulated processes, allocate simulated memory, and view basic system information.

This manual is intended for ordinary users and explains how to prepare the program, start it, log in, use its commands, handle common errors, and exit safely.

Important:
File operations affect files inside the Mini OS `mini_os_files` workspace, not the computer's general file system. Process and memory information are simulated and reset when the program exits.

2. **System Requirements**

Requirement	Minimum / Notes
Python	Python 3.9 or newer
Operating environment	Windows Command Prompt (CMD); PowerShell or another terminal may also work
Project files	Extracted NeOS project folder containing `main.py`, `users.py`, `filesystem.py`, `processes.py`, and `memory.py`
Dependencies	No third-party packages required; MiniOS uses Python's standard library
Storage	Enough disk space to store the project and files created in the MiniOS workspace

3. **Installation**

1. Download or copy the supplied `NeOS.zip` project archive to your computer.
2. Extract the ZIP file. Open the extracted `NeOS` folder and confirm that `main.py` is inside it.

3. Check that Python is installed. Open Command Prompt and run:
`python --version`
If this command is not recognized, install Python 3.9 or newer and enable the option to add Python to PATH. On some systems, use `py --version` and the `py` launcher instead.

4. No separate package installation is needed. MiniOS uses only the Python standard library.

5. Keep the project files together in the same folder. MiniOS creates or updates `users.json` and the `mini_os_files` workspace alongside the program.

4. How to Start MiniOS

1. Open Command Prompt (CMD) or PowerShell.

2. Change to the extracted project folder. Replace the example path with the actual location on your computer.

```
cd path\to\NeOS
```

3. Start the program:

```
python main.py
```

If your computer uses the Python launcher, you can try `py main.py`. The MiniOS title and login prompt should appear. Type commands at the prompt after logging in.

Example prompt:

```
student@MiniOS:~>
```

5. How to Login

1. Type a selected demo account for the username and password.

6. Command List

Command Name	Purpose	Syntax	Example
help	Displays Available Commands	help	help
clear	Clears the terminal screen.	clear	clear
date	Displays the computer's current date in YYYY-MM-DD.	date	date
time	Displays the current time in HH:MM:SS format.	time	time
whoami	Displays the username currently signed in.	whoami	whoami
users	Lists registered usernames available in the local MiniOS user file.	users	users
login	Displays a message explaining that signing out and back in is done through the logout flow; it does not directly ask for credentials.	login	login
logout	Signs out the current user and starts the login process again.	logout	logout
history	Shows commands entered during the current program session.	history	history

uptime	Shows how long MiniIOS has been running, in hours, minutes, and seconds.	uptime	uptime
systeminfo	Displays the MiniOS name and version, architecture, simulated CPU, simulated memory, current use, and workspace name.	systeminfo	systeminfo
exit	Shuts down MiniOS and prints a goodbye message.	exit	exit
dir	Lists the files and directories in the current MiniOS location. ls is an alias for dir.	dir	dir
mkdir	Creates a new directory in the current location.	mkdir	mkdir documents
create	Creates an empty file.	create	create notes.txt
read	Displays the contents of a file.	read	read notes.txt
write	Writes text to a file. Put the text in quotation marks if it contains spaces.	write	write notes.txt "Operating Systems Project”
delete	Deletes a file.	delete	delete notes.txt
rename	Renames a file or directory.	rename	rename notes.txt class-notes.txt
cd	Changes the current directory. Use cd.. to move to the parent directory.	cd	cd documents cd..

pwd	Displays the current MiniOS directory path.	pwd	pwd
process	Lists the simulated processes currently managed by MiniOS.	process	process
run	Starts a simulated process and assigns it a process ID (PID).	run	run calculator
kill	Terminates a simulated process using its PID, as shown in the process list.	kill	kill 1
memory	Displays total, used, and free simulated memory.	memory	memory
allocate	Allocates the requested amount of simulated memory in megabytes and reports an allocation ID.	allocate	allocate 200
free	Releases a previous simulated memory allocation using its allocation ID.	free	free 1

7. Error Handling

- Mkdir existing document

Error: [WinError 183] Cannot create a file when that file already exists: 'C:\\NeOS\\mini_os_files\\document'

- Delete non-existing file

Error: File does not exist.

- Empty username & password

Invalid username or password. Please try again.

- Kill non-existing ID

Invalid username or password. Please try again.

- Allocate more than maximum memory

Invalid username or password. Please try again.

- Allocate instead of numbers; you type letters

Error: Memory amount must be a whole number.

8. How to exit NeOS

1. Type “exit” to close the program.